SIMATS ENGINEERING

ASSESSMENT TOOL 2 – Concept Mapping

COURSE NAME: Cryptography and Network Security
COURSE CODE: CS A5103

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks: 100**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I RAKSHITHA R R (192324156) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Rakshitha R R

Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
   a. Show how each attack leads to specific threats and risks
   b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
   a. Show differences and similarities
      i. Map how each affects the Data structure and Security level
      ii. Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
   a. Show relationships between these concepts
   b. Analyze how each contributes to secure communication
   c. Include one real-world application for each

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
   a. Show how each concept is mathematically connected
   b. Map their role in encryption/decryption
   c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	*15*	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	*10*	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

CSA5103

CRYPTOGRAPHY & NETWORK SECURITY

CO1 - AT-2

CONCEPT MAPPING

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CONCEPT MAP -1 NUMBER THEORY

Euler's Phi function $\phi(n)$
Counts the integers upto n that are relative prime to n.

ϕ

Used as Modulus for

Key RSA STEP:
$\phi(n) = (p-1)(q-1)$

Is the Foundation for

$\varphi \bmod 1$

Special Case when modulus is a prime (p)

Euler's Theorem: $a^{\phi n} \equiv 1 \pmod n$
The mathematical "trapdoor" proof. If a and n are coprime; $a^{\phi(n)} \equiv 1 \pmod n$.
*** MOST CRITICAL FOR PUBLIC KEY CRYPTOGRAPHY ***
Role: Guarantees in applying the private key 'd', perfectly reverses the public key 'e' to recover the original text message (m).

$(m^e)^d \equiv m \pmod n$

$\bmod \equiv$

Extended Euclidean Algorithm (EEA)
An efficient algorithm to find the Greatest Common divisor GCD of two non-integers and express it as a linear combination.
$ax + by = \gcd(a, b)$

$ax + b$

Efficiently Computes the d parameter given e and $\phi(n)$.

provides theoretical Basis

$e \cdot d = \varphi(n)$

Fermat's Little Theorem:
$a^{(p-1)} \equiv 1 \pmod p$
A special Case of Euler's Theorem where n is a prime p and $\phi(n) = p-1$.
Role: Heavily used for fast primality Testing (eg miller Rabin) to securely generate the massive prime numbers 'p and q' needed for RSA

Modular Inverse:
$e \cdot d \equiv 1 \pmod{\phi(n)}$
The mathematical relationship where decryption key 'd' is the multiplicative inverse of encryption key 'e' modulo $\phi(n)$.
Role: Critical for RSA key Generation to Calculate the private key decryption 'd' so that $d = e^{(-1)} \bmod \phi(n)$.

S.No: 1

CONCEPT MAP - 2 SECURITY GOALS, ATTACKS, THREATS, & RISKS

[SECURITY GOALS & MITIGATION OF THREATS/RISKS

